

AGENTIC AI · PRIVACY-AWARE · HIGH-UTILITY SYNTHETIC DATA

Agentic AI Based Synthetic Data Generator

An Autonomous AI System for Intelligent, Privacy-Aware and High-Utility Synthetic Data Generation



Python · LLM · LangGraph · SDV · Pandas · NumPy · Scikit-learn

PROJECT TECHNICAL PRESENTATION · VIVA & EVALUATION

Why Synthetic Data Generation Is Still Difficult

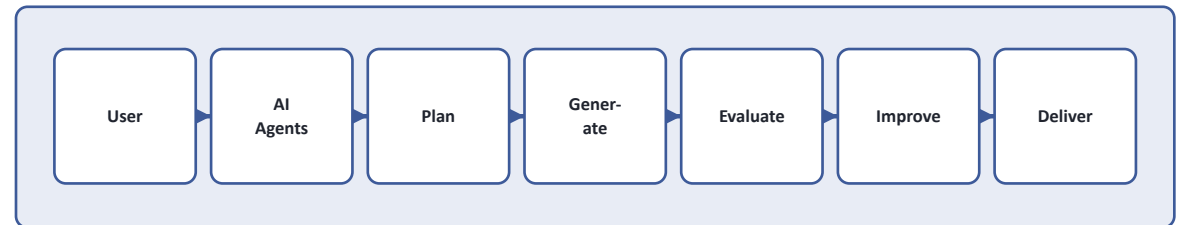
KEY CHALLENGES

- Manual schema design and configuration
- Requires deep technical expertise
- Difficult to preserve relationships between columns
- Synthetic data may not match real distributions
- Potential privacy leakage
- Poor downstream ML utility
- Manual, ad-hoc quality evaluation
- Repeated, manual parameter tuning

TRADITIONAL APPROACH



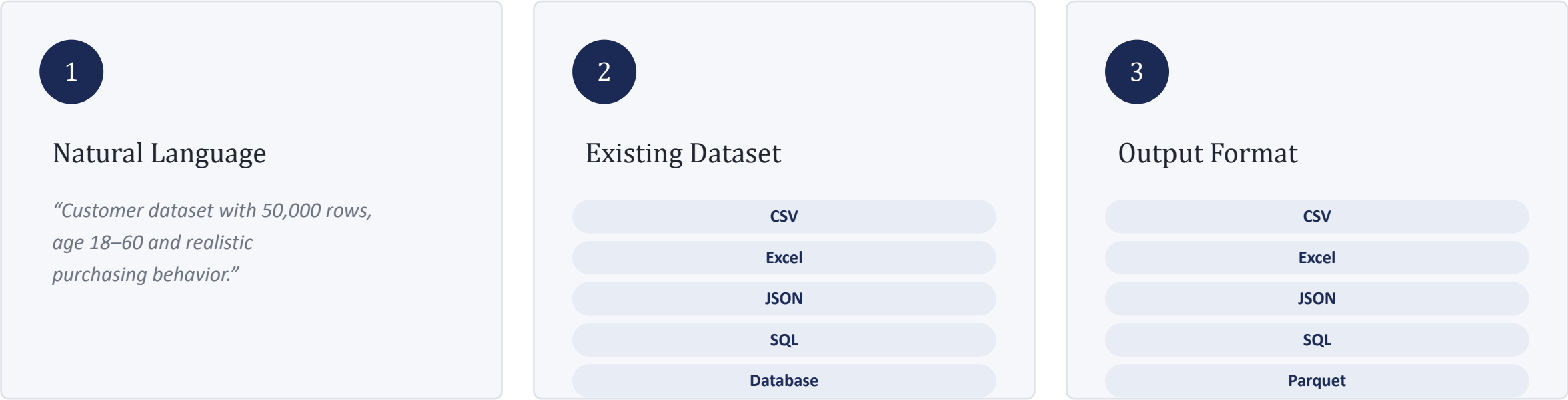
PROPOSED APPROACH



The problem is not only generating data —
it is generating data that is useful, realistic and safe.

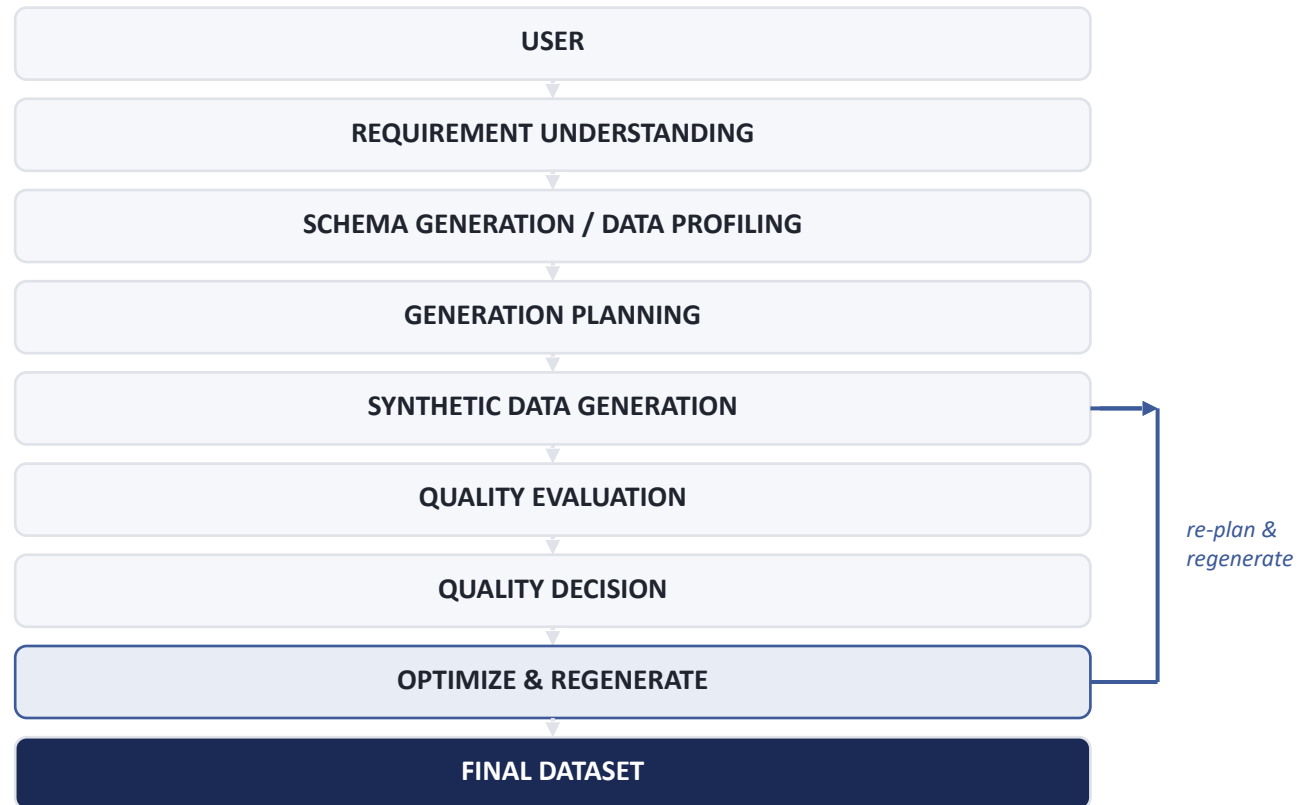
What Are We Building?

An agentic AI platform that converts natural-language dataset requirements or existing datasets into high-quality synthetic data through automated schema generation, profiling, generation planning, evaluation and iterative optimization.



The user specifies WHAT they need — the system determines HOW to create it.

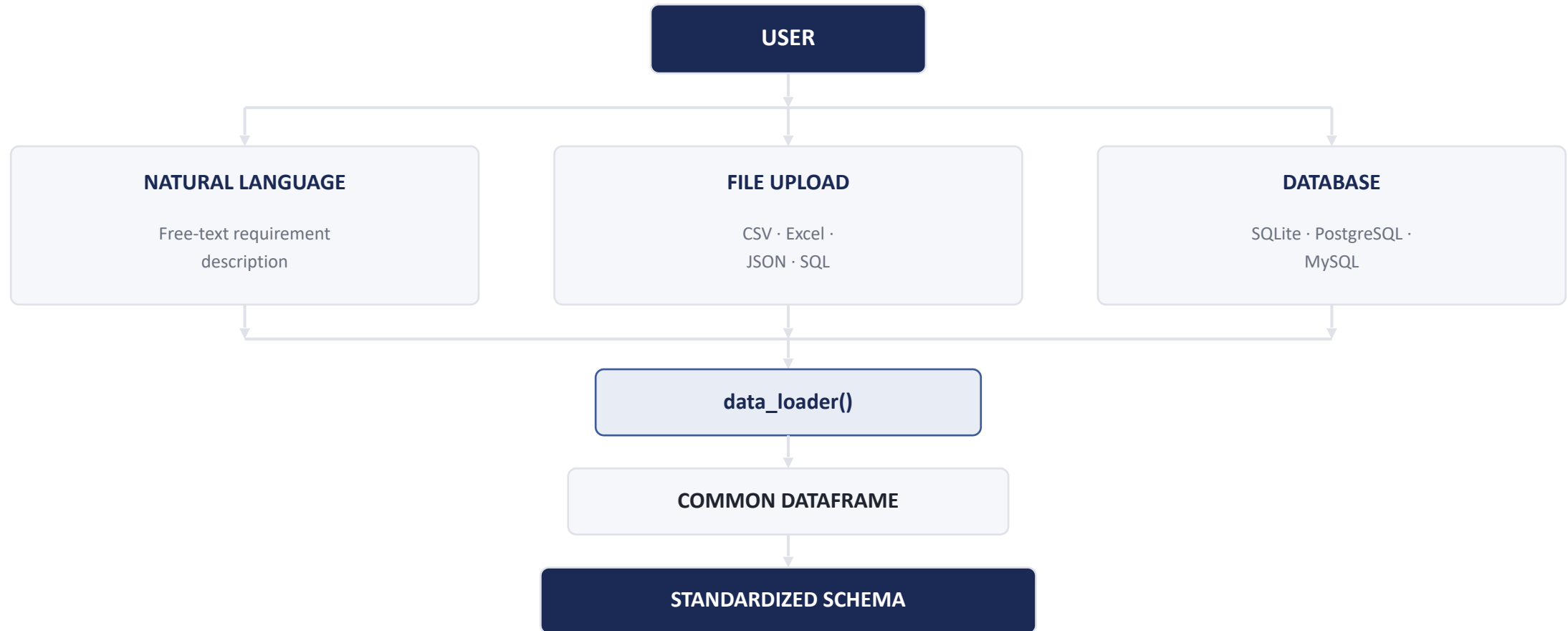
From Requirement to Final Dataset



Observe → Reason → Plan → Act → Evaluate → Improve

This is the fundamental agentic behavior.

Flexible Data Ingestion



Input Format ≠ Internal Representation

The Agents

01

Requirement & Schema Agent

Understands user requirements and creates/infers a structured schema.

02

Profiling Agent

Understands existing data — quality, distributions, missing values, correlations.

03

Generation Planner Agent

Decides generation strategy and configuration from schema and profile.

04

Generation Agent

Executes the plan — invokes SDV and returns the synthetic DataFrame.

05

Evaluation Agent

Determines acceptability — statistical quality, privacy, ML utility.

06

Optimization Agent

Analyzes failures and modifies configuration to trigger regeneration.

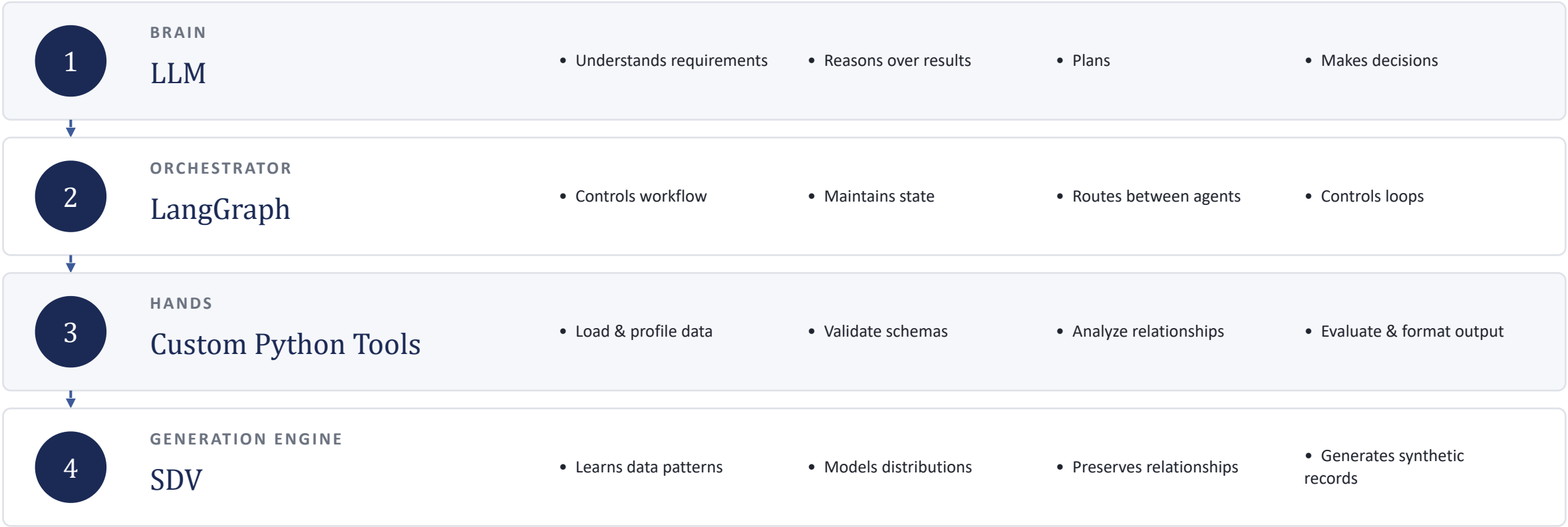
LangGraph orchestrates all six agents.

Tools We Build Ourselves



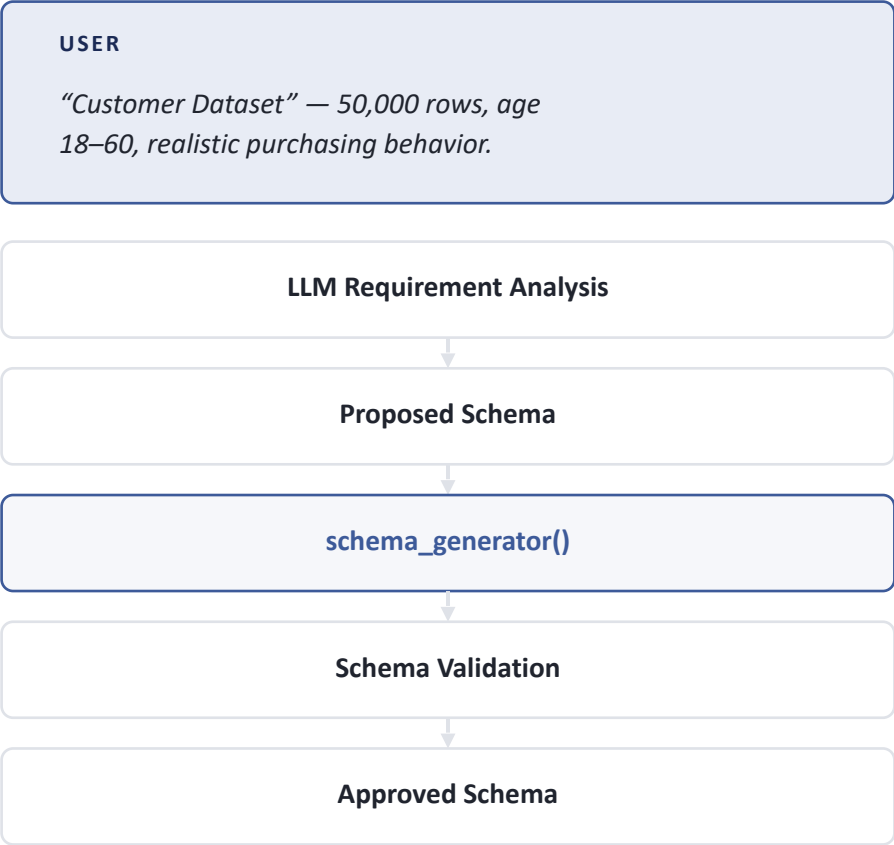
These are our custom Python tools — not separate AI models.

How the Technologies Work Together



LLM thinks. LangGraph orchestrates. Python tools act. SDV generates.

From Natural Language to Structured Schema



EXAMPLE SCHEMA

Column	Type
customer_id	ID
age	Numerical
gender	Categorical
city	Categorical
income	Numerical
signup_date	Datetime
total_orders	Numerical
total_spent	Numerical
customer_segment	Categorical

LLM handles semantic understanding. Python handles deterministic validation.

Understanding the Source Data

Data Quality

- Missing values
- Duplicates
- Invalid values
- Outliers

Statistics

- Mean · Median
- Variance
- Min / Max
- Distribution

Categorical Analysis

- Unique values
- Cardinality
- Category proportions

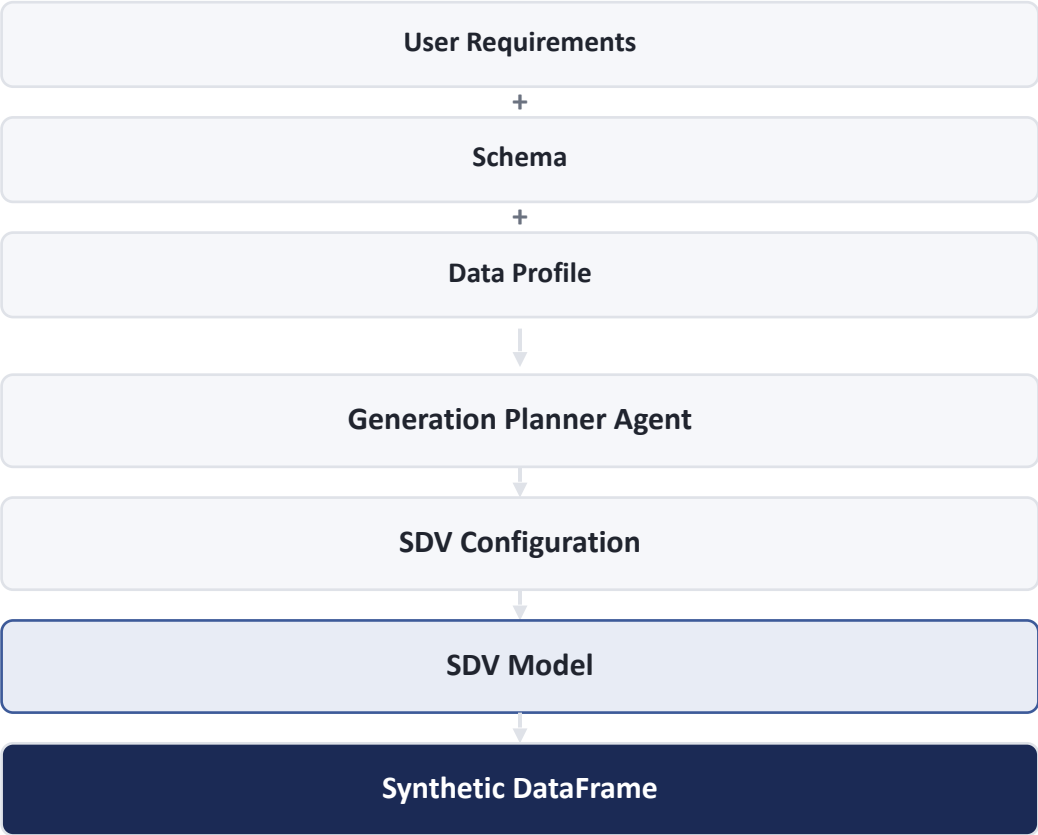
Relationships

- Correlation
- Dependencies
- Feature interactions

Example relationship: `total_spent ≈ total_orders × average_order_value`

Pandas · NumPy · dataset_profiler() · relationship_analyzer()

How SDV Generates the Data



SDV APPROACHES

Gaussian Copula

Models statistical relationships between columns.

CTGAN

Suited to complex, non-linear tabular distributions.

TVAE

Neural-network-based tabular synthesis approach.

LLM does NOT generate the rows.

LLM → decides / plans
SDV → learns patterns & generates records

Is the Synthetic Data Actually Good?

1

Statistical Quality

- Distribution, mean, median, variance
- Category proportions
- Correlations & feature relationships
- Missing-value patterns

2

Privacy

- Exact duplicates
- Highly similar records
- Potential memorization
- Re-identification risk indicators

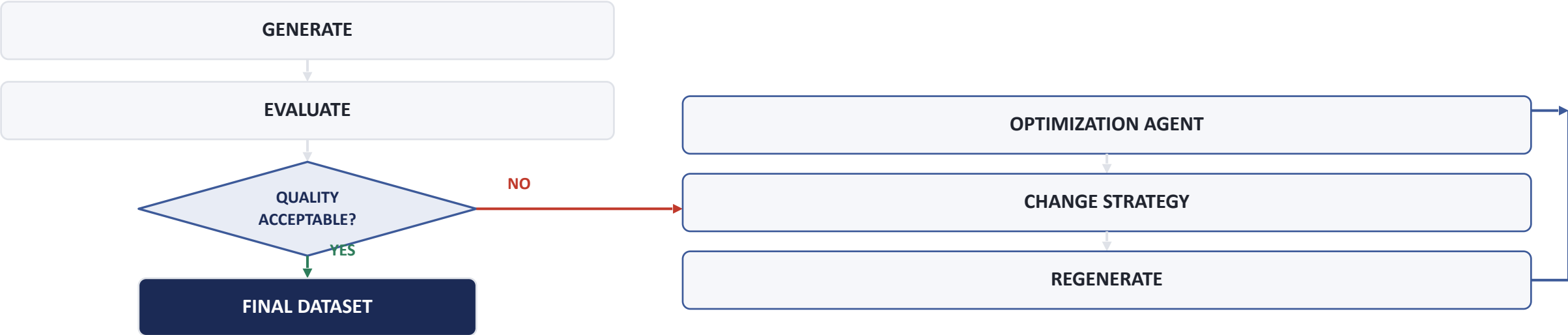
3

ML Utility

- Train same model on real vs synthetic
- Classification: Accuracy, Precision, Recall, F1, ROC-AUC
- Regression: MAE, RMSE, R^2

Good synthetic data should retain useful statistical structure and downstream utility — without unnecessarily exposing source records.

Generate → Evaluate → Improve



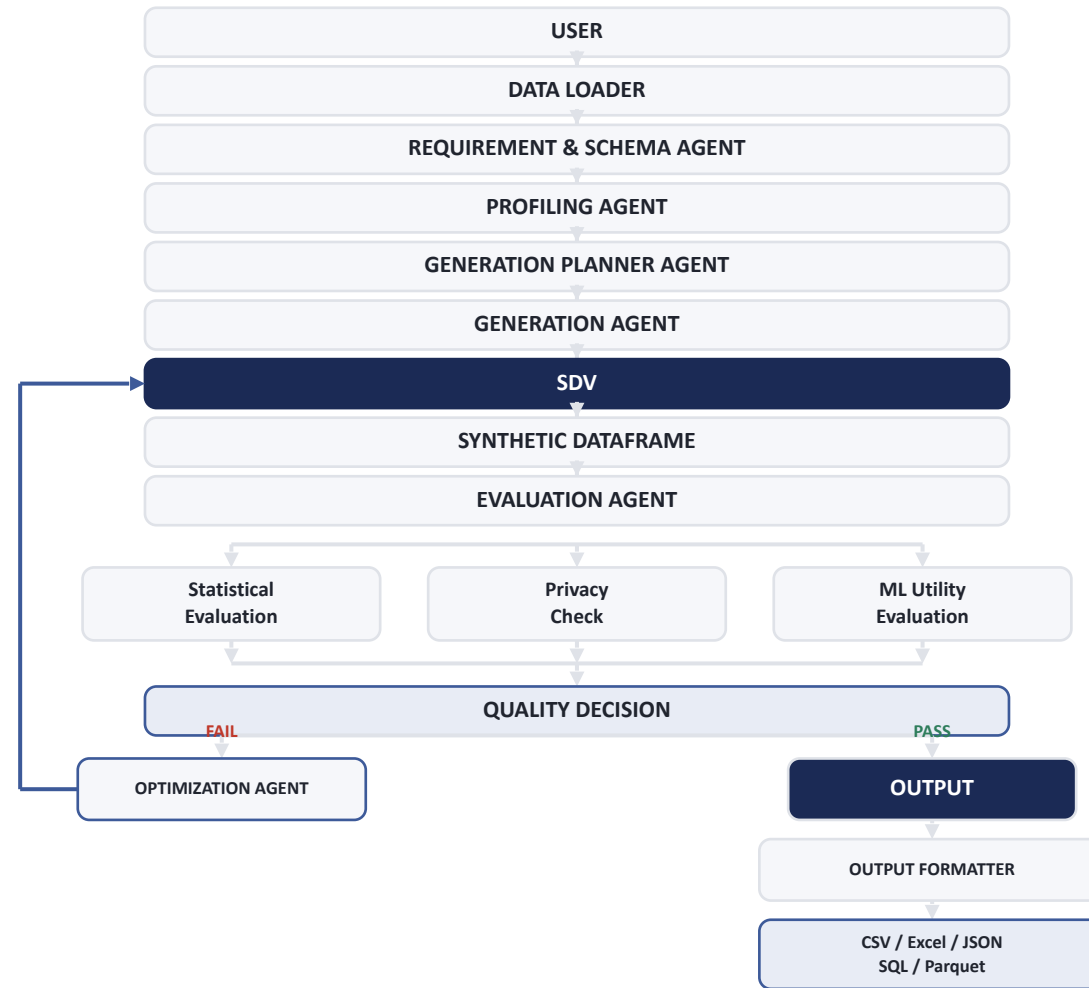
The Optimization Agent identifies why the result is weak and modifies the generation strategy before another SDV generation cycle.

QUALITY SCORE ACROSS ITERATIONS



This feedback loop is what makes the system agentic.

Complete System Architecture



Tech Stack & Applications

TECH STACK

Programming	Python
AI	LLM API
Agent Orchestration	LangGraph
Synthetic Data	SDV
Data Processing	Pandas · NumPy
Machine Learning	Scikit-learn
Visualization	Matplotlib · Seaborn
Interface	Streamlit

APPLICATIONS

Healthcare

Synthetic patient datasets for development and testing.

Finance

Synthetic transaction and customer datasets.

E-commerce

Customer behavior and purchase datasets.

Cybersecurity

Synthetic security and event datasets.

Machine Learning

Safe training and experimentation datasets.

Privacy is treated as a design and validation objective — not a guaranteed or regulatory-certified outcome.

CONCLUSION

From Manual Data Generation to Autonomous Data Engineering

The Agentic AI Based Synthetic Data Generator transforms synthetic data generation from a manual configuration process into an autonomous workflow that understands requirements, analyzes data, plans generation, evaluates quality and continuously improves the result.

